"Bakery Management and Online Ordering Web Application"

A PROJECT REPORT

Submitted by: -

Name - _____

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In partial fulfillment for the award of the degree of

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Department of UIC

Chandigarh University

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BONAFIDE CERTIFICATE

Certified that this project report titled "Bakery Management and Online Ordering Web Application" is the bonafide work carried out under supervision in partial fulfillment for the award of the degree of MCA at Chandigarh University.

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Submitted for the project viva-voce examination held on _____

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

TABLE OF CONTENT

CHAPTER 1. INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

1.2 Identification of Problem

1.3 Identification of Tasks

1.4 Timeline

CHAPTER 2. LITERATURE REVIEW / BACKGROUND STUDY

2.1 Timeline of the Reported Problem

2.2 Existing Solutions

2.3 Bibliometric Analysis

2.4 Review Summary

2.5 Problem Definition

2.6 Goals / Objectives

CHAPTER 3. DESIGN FLOW / PROCESS

3.1 Evaluation & Selection of Specifications / Features

3.2 Design Constraints

3.3 Analysis of Features and Finalization Subject to Constraints

3.4 Design Flow

3.5 Design Selection

3.6 Implementation Plan / Methodology

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

4.2 Testing and Validation

4.3 Performance Analysis

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

5.2 Future Work

REFERENCES

PLAGIARISM REPORT

CHAPTER 1

INTRODUCTION

1.1 Identification of Client / Need / Relevant Contemporary Issue

- Online bakery and food ordering is increasing due to smartphone and internet growth.
- Customers prefer quick menu browsing, digital ordering, and payment flexibility.
- Small bakery businesses need affordable digital systems for better order management.
- Users expect modern interfaces, order tracking, and payment options in one platform.
- Hence, a bakery management and ordering web application is required.

1.2 Identification of Problem

- Manual order taking causes mistakes and communication gaps.
- No centralized order history and payment status tracking in traditional process.
- Difficulty in handling peak-hour orders and repeated customer details.
- Limited digital visibility for menu categories and offers.
- Need for a structured, database-backed online ordering system.

1.3 Identification of Tasks

- Requirement analysis and feature finalization.
- UI/UX design and responsive frontend development.
- Cart logic, checkout workflow, and route integration.
- Backend API development with Node.js and MySQL.
- Payment integration (Razorpay and UPI QR).
- Testing, bug fixing, and report documentation.

1.4 Timeline

- Week 1 - Requirement gathering and architecture planning.
- Week 2 - UI design and frontend base implementation.
- Week 3-4 - Backend APIs and database integration.
- Week 5 - Payment modules, tracking, and order history features.
- Week 6 - Testing, alignment/theme refinements, and documentation.

CHAPTER 2

LITERATURE REVIEW / BACKGROUND STUDY

2.1 Timeline of the Reported Problem

- Before 2012 - Bakery and food orders mostly handled offline/phone-based.
- 2013-2018 - Mobile-first ordering platforms became mainstream.
- 2019-2022 - Contactless ordering and online payments accelerated adoption.
- Post 2022 - Local businesses need hybrid models with gateway + UPI methods.

2.2 Existing Solutions

- Major platforms provide menu browsing, cart, checkout, and order tracking.
- Payment methods include wallets, UPI, cards, and COD.
- Systems are scalable but often expensive for small independent bakeries.

2.3 Bibliometric Analysis

- Digital ordering improves convenience and conversion rates.
- Online payments improve transaction speed but require secure verification.
- Large marketplaces provide visibility but charge high commissions.
- Simple custom web systems are preferred for academic and SME use cases.

2.4 Review Summary

- Existing solutions are feature-rich and technically mature.
- There is clear need for a simplified and customizable bakery web system.
- A full-stack academic project can cover core functionality effectively.

2.5 Problem Definition

- Develop a user-friendly bakery ordering platform with responsive UI.
- Support product browsing, cart management, and checkout.
- Store order data in MySQL with payment status visibility.
- Provide order tracking and history retrieval modules.

2.6 Goals / Objectives

- Build complete frontend and backend integration.
- Implement COD, Razorpay, and QR payment workflows.
- Enable order tracking by ID and order history by contact details.
- Ensure maintainable code structure for future enhancements.

CHAPTER 3

DESIGN FLOW / PROCESS

3.1 Evaluation & Selection of Specifications / Features

- Responsive homepage with category-wise food menu.
- Dynamic add/remove cart and total calculation.
- Checkout form with delivery details.
- Payment methods: COD, Razorpay, QR code.
- Track Order page and My Orders history page.
- Theme switch (light/dark) and improved UI alignment.

3.2 Design Constraints

- Academic timeline and limited development window.
- Need to keep architecture simple and modular.
- QR payment manual confirmation in current phase.
- Production-grade auth/admin modules out of current scope.

3.3 Analysis of Features and Finalization Subject to Constraints

- Focused on core customer ordering flow first.
- Added backend-backed order tracking instead of complex real-time logistics.
- Implemented essential payment methods without over-engineering.
- Enhanced UI after functional stability was achieved.

3.4 Design Flow

- User opens app and explores menu categories.
- User adds products to cart and reviews totals.
- User fills delivery information on checkout page.
- User chooses payment mode and confirms order.
- System stores order and allows tracking/history lookup.

3.5 Design Selection

- Compared static UI-only design vs full-stack transactional design.
- Selected full-stack approach for realistic implementation and learning outcomes.
- Chosen architecture supports future migration to enterprise modules.

3.6 Implementation Plan / Methodology

- React + Vite for frontend module development.
- Node.js + Express for API and business logic.
- MySQL schema for persistent transactional storage.
- Iterative testing, bug fixing, and UX refinement cycle.

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

- Developed full-stack bakery ordering web application.
- Integrated cart, checkout, payment, and order tracking modules.
- Connected backend APIs to MySQL database tables (orders, order_items).
- Implemented My Orders page with reorder action and status badges.

4.2 Testing and Validation

- Validated cart add/remove and total calculation logic.
- Tested form validation and required field handling.
- Tested COD, Razorpay, and QR payment paths.
- Tested order tracking endpoint and history retrieval endpoint.
- Validated responsive behavior and theme visibility improvements.

4.3 Performance Analysis

- Fast UI interactions due to component-based rendering.
- Stable API response for core operations in local environment.
- Improved usability with aligned sections and clear navigation.
- Project meets academic objectives with practical feature set.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

- Successfully developed bakery management and online ordering solution.
- Addressed manual ordering limitations through digital workflow.
- Implemented multi-payment options and order visibility features.
- Achieved project objectives in functionality, usability, and modular design.

5.2 Future Work

- Add authentication and role-based admin dashboard.
- Integrate automated QR payment confirmation workflow.
- Add delivery status pipeline and notification system.
- Deploy to production cloud infrastructure with CI/CD.
- Add analytics and recommendation modules.

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PLAGIARISM REPORT

Project Title: Bakery Management and Online Ordering Web Application

Student Name: _____

UID: _____

Course: Master of Computer Application (MCA)

University: Chandigarh University

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I hereby declare that this project report is my original work. All sources used in this report are properly cited in the references section.

Signature of Student: _____

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